PROGRAM

```
#include <stdio.h>

struct Process {
    int pid;
    int bt;
    int rem_bt; // Remaining Burst Time
    int wt;
    int tat;
};

int main() {
    int n, qt;
    int count = 0; // Completed process count
    int total_time = 0; // Global time tracker
    float total_wt = 0, total_tat = 0;

    printf("Enter the number of processes\n");
    scanf("%d", &n);

    struct Process p[n];

    for(int i=0; i<n; i++) {
        p[i].pid = i + 1; // Manual uses P1, P2 etc.
        printf("Enter the burst time of the process%d ", i+1);
        scanf("%d", &p[i].bt);

        // Manual Output check: Invalid Input [cite: 690]
        if(p[i].bt < 0) {
            printf("Invalid Input\n");
            return 0;
        }
        p[i].rem_bt = p[i].bt;
    }

    printf("Enter the quantum time\n");
    scanf("%d", &qt);

    // Step 6: Infinite Loop
    while(1) {
        count = 0; // Reset completed count for this pass check

        for(int i=0; i<n; i++) {
            if(p[i].rem_bt == 0) {
                count++;
                continue;
            }

            int exec_time;

            if(p[i].rem_bt > qt) {
                exec_time = qt;
                p[i].rem_bt -= qt;
            } else {
                exec_time = p[i].rem_bt;
```

```
            p[i].rem_bt = 0;
        }

        total_time += exec_time;

        // If process just finished, set TAT
        if(p[i].rem_bt == 0) {
            p[i].tat = total_time;
        }
    }

    // Step 6.3: Check if all finished
    if(count == n) break;
}

// Step 7: Calculate Waiting Time
printf("\nProcess\tburst time\twaiting time\tturnaround time\n");
for(int i=0; i<n; i++) {
    p[i].wt = p[i].tat - p[i].bt;
    total_wt += p[i].wt;
    total_tat += p[i].tat;

    printf("P%d\t%d\t\t%d\t\t%d\n", p[i].pid, p[i].bt, p[i].wt, p[i].tat);
}

printf("\nAverage waiting time = %f", total_wt / n);
printf("\nAverage turn around time = %f\n", total_tat / n);

return 0;
}
```

OUTPUT

OUTPUT 1

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_3$ gcc 3d.c
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_3$ ./a.out
Enter the number of processes
4
Enter the burst time of the process1 8
Enter the burst time of the process2 6
Enter the burst time of the process3 3
Enter the burst time of the process4 5
Enter the quantum time
2

Process burst time      waiting time    turnaround time
P1      8               14              22
P2      6               13              19
P3      3               10              13
P4      5               15              20

Average waiting time = 13.000000
Average turn around time = 18.500000
```

OUTPUT 2

```
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_3$ ./a.out
Enter the number of processes
4
Enter the burst time of the process1 6
Enter the burst time of the process2 -8
Invalid Input
blackie@blackie-pc:~/GEC_PKD_IT/S4/OS_LAB/EXPERIMENT_3$
```